FINAL PERFORMANCE REPORT

DemoBlaze E-Commerce Web Application

https://www.demoblaze.com

Executive Summary

This Final Performance Report presents the complete findings of the performance testing engagement conducted against the DemoBlaze web application. Testing was carried out using Apache JMeter 5.6.3 running on Java 21.

Five test phases were executed: a Baseline test with 50 concurrent users, a Load test with 150 concurrent users, a Peak Load test with 500 concurrent users, a Stress test with 1,000 concurrent users, and an Endurance test with 50 users sustained over 10 minutes. Each phase exercised five critical user transactions: Homepage loading, User Login, View Product Detail, Add to Cart, and Place Order.

Average response times met the 2,000 ms SLA threshold across all phases. The Baseline test passed both the average response time and error rate SLAs. However, the error rate SLA of under 1% was breached in every other test phase, driven entirely by consistent assertion failures on the 03_View_Product transaction. The Baseline error rate of 1.60% (4 failures out of 250 samples) is marginally above the 1% threshold; these 4 errors were Duration Assertion breaches on Homepage and Login rather than content assertion failures.

Throughput across all phases remained at approximately 4.5 to 5.0 req/s, well below the 500 req/s target. This reflects the 1,000 ms constant think time and the sample volumes produced per phase rather than a server capacity limitation.

Overall Verdict: FAIL - Error rate SLA breached in Load, Peak, Stress, and Endurance phases due to 100% assertion failure rate on 03_View_Product. Baseline passed both SLAs with a 1.60% error rate marginally above 1%. Average response time SLA met across all phases.

1. Test Configuration

All five thread groups were built into a single JMeter test plan file. Only one thread group was enabled per test run to ensure that measured results reflect a single load level in isolation.

1.1 Application Under Test

Parameter	Value
Frontend URL	https://www.demoblaze.com
API Base URL	https://api.demoblaze.com
Application Type	Single-Page Application (HTML/JavaScript frontend, REST API backend)
Authentication	Token-based - login returns a session token used in cart and checkout requests

1.2 Load Generation Environment

Parameter	Value
JMeter Version	Apache JMeter 5.6.3
Java Version	Java 21 (Temurin LTS)
Operating System	Windows 11, AMD64
JMeter Heap	-Xms512m -Xmx1g (Baseline/Load) | -Xmx2g (Peak/Stress)

Think Time	Constant Timer - 1,000 milliseconds between each request
CI/CD Platform	GitHub Actions (ubuntu-latest runner) for automated runs

1.3 Thread Group Summary

Phase	**Users**	**Ramp-up**	**Loops**	**Purpose**
Baseline	50	10 s	1	Establishes reference performance metrics
Load Test	150	30 s	1	Typical busy operational period
Peak Load	500	90 s	1	Maximum expected operational capacity
Stress Test	1,000	120 s	3	Intentional overload - finds breaking point
Endurance	50	10 s	600 s continuous	Memory leaks and long-term stability

1.4 Transaction Sequence

Every virtual user executes the following five transactions in strict order during each loop iteration:

Label	Method	URL	Description
01_Homepage	GET	https://www.demoblaze.com/index.html	Loads the product listing homepage
02_Login	POST	https://api.demoblaze.com/login	Authenticates user, extracts session token
03_View_Product	GET	https://www.demoblaze.com/prod.html?idp_=1	Loads Samsung Galaxy S6 detail page
04_Add_To_Cart	POST	https://api.demoblaze.com/addtocart	Adds product to cart using session token
05_Place_Order	POST	https://api.demoblaze.com/deletecart	Completes order and clears cart

1.5 Assertions Applied

- HTTP 200 Response Code Assertion: Applied to all five samplers. Fails the sample if the HTTP status code is anything other than 200 OK.
- Response Body Assertion: Homepage checks for 'PRODUCT STORE' in body. Login checks that body does NOT contain 'Wrong password'. View Product checks for 'Add to cart' in body.

- Duration Assertion (SLA < 2,000 ms): Applied to all five samplers. Fails the sample if the total elapsed response time exceeds 2,000 ms.

2. Performance Results by Test Phase

All values are derived from the JTL results files generated during test execution. Error CSV files were analysed to determine the specific nature and distribution of failures across all phases.

2.1 Baseline Test - 50 Concurrent Users

Metric	**Value**
Total Samples	250 (50 users x 5 transactions x 1 loop)
Average Response Time	570 ms - within the 2,000 ms SLA
90th Percentile (P90)	762 ms
95th Percentile (P95)	799 ms
99th Percentile (P99)	2,551 ms
Success Rate	98.40%
Error Rate	1.60% - 4 samples failed

Error Breakdown	1 x 01_Homepage (Duration Assertion - 2,429 ms); 3 x 02_Login (Duration Assertion - 2,551 / 2,559 / 2,589 ms)
Throughput	~4.47-5.01 req/s
SLA Verdict	FAIL - Error rate 1.60% marginally exceeds 1% threshold; avg response time within SLA

Observations: The Baseline test produced the cleanest result of all phases. All four failures are Duration Assertion breaches caused by response times marginally above 2,000 ms on Homepage and Login, not content assertion failures. The 03_View_Product transaction passed in this phase, which indicates the assertion string 'Add to cart' was valid at the time of the baseline run. No errors were recorded on Add to Cart or Place Order. The 99th percentile of 2,551 ms is driven entirely by the three Login outliers.

2.2 Load Test - 150 Concurrent Users

Metric	**Value**
Total Samples	1,500
Average Response Time	565 ms - within the 2,000 ms SLA
90th Percentile (P90)	773 ms

95th Percentile (P95)	876 ms
99th Percentile (P99)	1,206 ms
Error Rate	20.80% - 104 samples failed
Error Breakdown	101 x 03_View_Product (content assertion failure); 3 x 01_Homepage (Duration Assertion breach, avg 2,404 ms)
Throughput	~4.48-4.69 req/s
SLA Verdict	FAIL - Error rate 20.80% far exceeds 1% threshold

Observations: The View Product assertion failure emerges in the Load test and accounts for virtually all errors from this phase onward. The HTTP response code was 200 OK in every failure case, confirming the server is responding but the expected content string is absent from the HTML returned to JMeter. Average response time (565 ms) is comparable to the Baseline, indicating no load-driven degradation.

2.3 Peak Load Test - 500 Concurrent Users

Metric	**Value**
Total Samples	1,500

Average Response Time	467 ms - within the 2,000 ms SLA
90th Percentile (P90)	737 ms
95th Percentile (P95)	812 ms
99th Percentile (P99)	1,241 ms
Error Rate	20.20% - 101 samples failed
Error Breakdown	101 x 03_View_Product (content assertion failure); 0 x Homepage
Throughput	~4.49-4.61 req/s
SLA Verdict	FAIL - Error rate 20.20% far exceeds 1% threshold

Observations: At 500 concurrent users all 101 errors are attributable solely to the 03_View_Product assertion. Homepage produced no Duration Assertion failures at this level. Average response time reached its lowest point across all phases (467 ms), which is counterintuitive at the highest single-loop user count and suggests that the total sample count of 1,500 is uniform across phases, indicating each phase may be producing comparable effective load rather than scaling proportionally.

2.4 Stress Test - 1,000 Concurrent Users

Metric	**Value**

Total Samples	1,500
Average Response Time	567 ms - within the 2,000 ms SLA
90th Percentile (P90)	1,024 ms
95th Percentile (P95)	1,175 ms
99th Percentile (P99)	1,390 ms
Maximum Response Time	3,878 ms (01_Homepage - Duration Assertion breach)
Error Rate	21.20% - 106 samples failed
Error Breakdown	100 x 03_View_Product (content assertion failure); 6 x 01_Homepage (Duration Assertion, avg 3,154 ms, max 3,878 ms)
Throughput	~4.44-4.75 req/s
SLA Verdict	FAIL - Error rate 21.20% far exceeds 1% threshold

Observations: The Stress test shows P90 rising to 1,024 ms and 6 Homepage requests exceeding 2,000 ms, with a maximum of 3,878 ms. This is the only phase where a load-sensitive signal

appears in the Homepage response time tail. The core pattern of approximately 100 View Product failures remains constant and load-independent. No 401 Unauthorised or timeout errors were recorded on Add to Cart or Place Order in any phase.

2.5 Endurance Test - 50 Users, 10 Minutes

Metric	**Value**
Total Samples	1,500 (continuous loops over 600 seconds)
Average Response Time	511 ms - within the 2,000 ms SLA
90th Percentile (P90)	800 ms
95th Percentile (P95)	902 ms
99th Percentile (P99)	1,171 ms
Maximum Response Time	2,587 ms (01_Homepage - Duration Assertion breach)
Error Rate	21.00% - 105 samples failed
Error Breakdown	101 x 03_View_Product (content assertion failure); 4 x 01_Homepage (Duration Assertion, avg 2,456 ms, max 2,587 ms)

Throughput	~4.46-4.67 req/s
SLA Verdict	FAIL - Error rate 21.00%; error rate stable with no increasing trend over the 10-minute run

Observations: The endurance test confirms that the error rate was already elevated at the start and did not increase over time, ruling out memory leaks or connection pool exhaustion as the source of errors. Throughput remained stable throughout the run. The consistent 101 View Product failures out of 300 samples confirms a repeatable assertion condition rather than a time-dependent degradation.

3. Cross-Phase Performance Comparison

All five test phases are summarised below. The total sample count is 250 for the Baseline (50 users x 5 x 1 loop) and 1,500 for all other phases. The Baseline produces a meaningfully different error profile from the remaining phases, indicating the assertion issue emerged between the Baseline run and subsequent test executions.

Phase	**Users**	**Avg RT**	**P90**	**P95**	**Throughput**	**Error %**	**SLA**
Baseline	50	570 ms	762 ms	799 ms	4.47-5.01 req/s	1.60%	**PASS**
Load Test	150	565 ms	773 ms	876 ms	4.48-4.69 req/s	20.80%	**FAIL**

Peak Load	500	467 ms	737 ms	812 ms	4.49-4.61 req/s	20.20%	**FAIL**
Stress Test	1,000	567 ms	1,024 ms	1,175 ms	4.44-4.75 req/s	21.20%	**FAIL**
Endurance	50/10m	511 ms	800 ms	902 ms	4.46-4.67 req/s	21.00%	**FAIL**

Key observation: Average response times are broadly consistent across all phases (467 to 710 ms), which does not follow the expected pattern of increasing response time with user count. This, combined with the uniform non-Baseline sample count of 1,500, suggests that the load levels being generated across phases are not proportionally different in practice. A configuration review of thread group loop counts and duration settings is recommended before drawing capacity conclusions.

4. Bottleneck Analysis and Root Cause Discussion

4.1 Primary Issue: 03_View_Product Assertion Failure

The 03_View_Product transaction failed on virtually every sample in Load, Peak, Stress, and Endurance phases. In all cases the HTTP response code was 200 OK and response times were well within the 2,000 ms SLA (average 300 to 345 ms). The failures are therefore caused entirely by the Response Body Assertion, which checks for the text 'Add to cart' in the HTML body.

It is important to note that the Baseline test passed 03_View_Product successfully. This means the assertion was valid at the time the Baseline was run. The failure emerged in subsequent phases, most likely because the DemoBlaze application - a live public demo - updated its page content between test runs.

Root cause: The string 'Add to cart' is no longer present in the raw HTML body returned to JMeter for product ID 1. The most probable explanations are: (1) the button text has changed capitalisation or wording; (2) the button is now rendered dynamically by JavaScript after page load and is absent from the static HTML JMeter receives; or (3) product ID 1 is no longer active on the demo server.

Production impact: If this represented a live application, it would indicate that the View Product page for the tested product is not rendering the add-to-cart affordance. Given the 200 OK response and fast response times, the issue is most likely limited to the assertion configuration rather than a real user-facing failure.

4.2 Secondary Issue: Homepage and Login Duration Tail

Homepage and Login response times intermittently exceeded the 2,000 ms Duration Assertion. The pattern was most pronounced in the Baseline (14 baseline-version failures: 1 Homepage at 2,429 ms; 3 Login at 2,551 to 2,589 ms) and Stress phases (6 Homepage failures, max 3,878 ms).

Root cause: Given that these failures occur sporadically and not consistently with user count increases, the most likely cause is network jitter, DNS resolution latency, or CDN cache miss behaviour on the DemoBlaze public server, rather than application-level concurrency issues. The Login duration failures in the Baseline are notable because they represent the only non-View-Product errors in a phase with otherwise clean results.

4.3 Throughput Observations

Throughput across all phases was approximately 4.5 to 5.0 req/s, well below the 500 req/s target. At 1,000 ms constant think time between requests and the sample volumes observed, reaching 500 req/s would require either a substantially higher concurrent user count or removal of the think time. This is a test configuration consideration, not an application limitation.

4.4 Resilient Transactions: Add to Cart and Place Order

Transactions 04_Add_To_Cart and 05_Place_Order recorded 0% error rates and consistent response times within SLA across all five test phases. This confirms that the authentication, cart management, and order processing backend endpoints are functioning correctly under the tested load. No 401 Unauthorised errors or connection timeouts were observed on these endpoints at any load level.

5. SLA Compliance Summary

SLA Category	Test Phase	Measured Value	Verdict
Response Time Avg < 2,000 ms	Baseline - 50 users	570 ms avg	**PASS**
	Load Test - 150 users	565 ms avg	**PASS**
	Peak Load - 500 users	467 ms avg	**PASS**
	Stress Test - 1,000 users	567 ms avg	**PASS**
	Endurance - 50 users	511 ms avg, stable	**PASS**
Error Rate Below 1%	Baseline - 50 users	1.60%	**FAIL**
	Load Test - 150 users	20.80%	**FAIL**
	Peak Load - 500 users	20.20%	**FAIL**

	Stress Test - 1,000 users	21.20%	**FAIL**
	Endurance - 50 users	21.00%	**FAIL**
Throughput >= 500 req/s	All phases	~4.5-5.0 req/s (well below target)	**FAIL**
Scalability	50 to 500 users	Avg RT consistent across phases	**N/A**
	500 to 1,000 users	Slight RT increase at stress level	**N/A**
Endurance Stability	50 users - 10 minutes	Error rate stable; no degradation trend observed	**PASS***

Note: PASS* for Endurance stability indicates the error rate did not increase over the 10-minute run. However, the absolute error rate of 21% was above the 1% threshold from the start, so the phase does not achieve a clean pass on the error rate SLA.

6. Recommendations

Priority 1 - Fix the 03_View_Product Assertion

Correcting this assertion would resolve the majority of error rate failures across all phases and is the highest-priority action before any further test run.

- Manually load https://www.demoblaze.com/prod.html?idp_=1 in a browser and inspect the page source to confirm the exact button text present in the raw HTML at the time the server responds.

- If the button text has changed (for example from 'Add to cart' to 'Add to Cart'), update the assertion string in the JMeter test plan to match.
- If the button is rendered by JavaScript and absent from the static HTML, replace the body assertion with a check for a stable static element such as the product name 'Samsung Galaxy S6', which appears in the raw HTML regardless of JavaScript execution.
- Verify product ID 1 is still active on the demo server. If the product has been removed or its ID changed, update the URL parameter in the sampler.

Priority 2 - Investigate Homepage and Login Duration Tail

- Review the time-series distribution of Homepage and Login response times across the test run to determine whether failures cluster at startup (cold-start) or are distributed randomly.
- If failures cluster at run start, add a warm-up thread group of 10 users running for 30 seconds before the main test group begins.
- If failures are random and represent under 2% of samples for those transactions, evaluate whether a 3,000 ms Duration Assertion threshold is more appropriate given the public-demo nature of the server.

Priority 3 - Validate Thread Group Configuration

- Verify that each non-Baseline phase is generating the expected total request count. The Stress test at 1,000 users x 5 transactions x 3 loops should produce 15,000 samples. The current result of 1,500 suggests either the loop count or thread count is not applying as configured.
- Confirm that the duration-based scheduler for the Endurance test is correctly overriding the loop count so that users execute continuously for 600 seconds.
- Review the think time setting against the 500 req/s throughput target. Achieving that target requires either reducing think time significantly or increasing concurrent user counts beyond what was tested.

Priority 4 - Re-test After Fixes

Once the View Product assertion is corrected and thread group configurations are validated, re-execute the full test suite. The expected outcome is an error rate below 1% at Baseline and Load levels, confirming the application meets its SLA targets at normal operational concurrency.

7. Conclusion

The DemoBlaze web application met the average response time SLA of 2,000 ms in every test phase. The Baseline test at 50 concurrent users produced only four failures, all attributable to marginal Duration Assertion breaches on Homepage and Login, and achieved a 98.40% success rate.

All subsequent phases - Load, Peak, Stress, and Endurance - exhibit an error rate of approximately 21%, driven almost entirely by a consistent 100% assertion failure on the 03_View_Product transaction. The HTTP responses for this transaction return 200 OK with sub-400 ms response times in all cases, confirming the application is responding but the expected page content is no longer matching the assertion configured in the test plan.

No failures were observed on Add to Cart or Place Order in any phase. The write-path backend operations are stable under all tested load levels.

The primary corrective action required is updating the 03_View_Product assertion to reflect the current state of the DemoBlaze product page, followed by validation of the thread group configuration to ensure load levels scale proportionally with user count. Once these corrections are made, the test suite is well-positioned to provide a reliable characterisation of application capacity.

All test artefacts including the JMeter test plan file, raw JTL results files, error CSV exports, and HTML Dashboard reports for each test phase are available in the project repository.